Sharp Robust QoS Certification from Six-Scalar Spectral Summaries

Fixed beamforming and RIS configurations under a partial-information model

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Abstract

Certifying that a fixed beamforming and reconfigurable intelligent surface (RIS) configuration satisfies quality-of-service constraints under a norm-bounded channel-state error is, with full geometry, a trust region problem solvable exactly. This paper asks what can be guaranteed when the certification interface retains only six scalar summaries of the nominal channel and the interference operator: desired and nominal amplitudes, desired-direction gain, orthogonal-subspace gain, and total spectral norm. Because these gains originate from one operator, they obey spectral compatibility that a comparison envelope using the gains separately ignores. We solve the resulting directional interference extremum in closed form — a continuous three-branch envelope — and obtain a safe worst-case QoS bound that dominates the separate-gain envelope under the same information. For the nondegenerate information class we prove the bound equals the infimum of the true worst-case margin over all instances sharing the six summaries, and exhibit an approximating construction sequence; at the degenerate boundary where the desired direction attains the total norm, a distinct phase-coupled envelope is derived and attained explicitly. Numerical checks reproduce the analytical envelopes; on the validation batch two independent exact formulations agree, and the certification gain over the separate-gain envelope is strict but modest and batch-dependent; the certificate is complementary to a quadratic bound that retains one additional scalar.

Keywords: robust QoS certification; partial information; information class; spectral compatibility; RIS; fixed configuration.

1 Introduction

After beamforming vectors and discrete RIS phases have been fixed for a candidate configuration, deployment, reuse, and screening still require verifying that every user's QoS constraint survives channel-state uncertainty. With the complete channel and interference geometry available, this verification is a ball-constrained quadratic problem that trust-region methods and the S-procedure solve exactly [2],[3],[4]; robust transmission design with imperfect CSI, including bounded-error RIS settings, is correspondingly mature [1],[4],[9]. The existence of exact full-information solvers is therefore not the issue this paper addresses.

In candidate screening over many configurations, when configuration descriptors are cached or exchanged across network components, or when a stored candidate is evaluated at several uncertainty radii, a compact fixed-size summary can be preferable to carrying the full quadratic geometry. If only six scalar summaries are retained — the desired-stream power, the nominal desired and interference amplitudes, and three gains of the interference operator — a precise question arises: what is the strongest worst-case QoS guarantee that is uniform over every instance consistent with these summaries? Envelopes that use the three gains separately ignore the fact that all three originate from one operator and therefore cannot attain their individual worst cases simultaneously.

We answer the question in three steps. First, we solve the directional interference extremum under the joint spectral constraints: given the desired-direction gain α, the orthogonal-subspace gain β, and the total spectral norm β_0, we evaluate the largest interference output that a parallel error amplitude x and an orthogonal budget y can jointly produce. The extremum is a continuous three-branch closed form proved directly from a two-by-two determinant argument, in the same spirit as classical contraction-completion and Buzano-type refined Cauchy–Schwarz inequalities [5],[6],[7]. Second, we embed the extremum into the worst-case QoS margin, obtaining a shared-error-budget lower envelope.

Third, we ask whether any safer certificate using the same six scalars can be strictly better, and we answer in the negative away from a degenerate spectral boundary.

The envelope is safe and dominates the separate-gain comparison envelope pointwise. Over the nondegenerate information class ($\beta_0>\alpha$) the envelope equals the infimum of the true worst-case margin over all instances with the same six summaries, and an explicit construction sequence approaches this infimum — a sharpness statement related to the information-based viewpoint of optimal recovery [8]. At the degenerate boundary $\beta_0=\alpha$, the equality $\beta_0=\alpha$ is itself one of the summaries; it forces the parallel and orthogonal interference outputs to be orthogonal, so the desired signal and the parallel interference share one phase within the boundary class, and the optimal envelope of that class is different and is attained by an explicit diagonal construction. The two envelopes can differ sharply for the same boundary summaries (0.59 versus −0.21 in a constructed example); the discontinuity is a property of the information classes, not of any physical channel.

The numerical study has five roles: verifying the analytical implementation to machine precision; exhibiting constructed instances where the new envelope certifies feasibility that the separate-gain envelope cannot; validating on preregistered, previously unused seeds, including a heterogeneous-link pool; showing that the six-scalar certificate and a quadratic bound retaining one additional scalar are complementary (no universal dominance, safe to combine by maximization); and cross-validating the certification statements of the validation batch with an independent S-procedure SDP formulation. The experiments support implementation correctness and complementary use; they do not prove the theorems, and the empirical acceptance gain over the separate-gain envelope is positive in the sensitivity, validation and heterogeneous-link batches, zero in a saturated easy batch, and modest wherever positive.

This work makes three contributions. First, we derive the spectral-compatible directional envelope: from the joint constraints imposed by the desired-direction gain, the orthogonal-subspace gain, and the total spectral norm of one interference operator, we obtain an explicit continuous three-branch extremum for the worst interference output under a shared error budget. Second, we establish information-class optimality: the induced shared-budget QoS envelope is safe, dominates the envelope that uses the three gains separately, equals the information-class infimum away from the degenerate boundary, and is complemented by a separate, explicitly attained phase-coupled envelope at the boundary $\beta_0=\alpha$. Third, we specify and validate a certification interface: exact trust-region and S-procedure references confirm the implementation on every validation problem, the envelope produces certifications the separate-gain envelope cannot, and, having no universal dominance relation with the one-additional-scalar quadratic bound, it serves as a complementary stage between cheap bounds and exact fallback.

2 Fixed configuration model, information class, and related work

2.1 Robust QoS margin

For a user, let $b \in \mathbb{C}^n$ be the nonzero desired beam, $C \in \mathbb{C}^{n\times m}$ the matrix of interfering beams, and $\hbar \in \mathbb{C}^n$ the nominal stacked channel. The actual channel is $\hbar + e$, with $\| e \| \leq r$. The SINR requirement is $\gamma > 0$, and $\sigma^2 \geq 0$ denotes noise power, not noise amplitude. We suppress user indices and define

$$q^*(r) = \min_{\|e\|\leq r} \{ |b^H(\hbar + e)|^2 - \gamma \| C^H(\hbar + e) \|^2 - \gamma\sigma^2 \} \quad (1)$$

All users satisfying $q_u{}^*(r_u) \geq 0$ makes the fixed candidate robustly feasible. A negative lower bound alone does not establish infeasibility.

2.2 Six scalar summaries and information-class nonemptiness

Set $v = b/\| b \|$ and $P = I - vv^H$. The retained summaries are

$$p = \| b \|^2 , \;\; a = |v^H\hbar|, \;\; I_0 = \| C^H\hbar \| \quad (2a)$$

$\alpha = \| C^H v \|$, $\beta = \| C^H P \|_2$, $\beta_0 = \| C \|_2$ (2b)

The spectral summaries necessarily satisfy

$\max\{\alpha, \beta\} \le \beta_0 \le \sqrt{\alpha^2 + \beta^2}$ (3)

Condition (3) alone does not characterize nonempty information classes, because extreme spectra force the nominal amplitudes.

Lemma 1 (nonemptiness). Fix $n \ge 5$, $m \ge 3$, $p > 0$ and a summary vector $s = (p, a, I_0, \alpha, \beta, \beta_0)$ with $a, I_0, \alpha, \beta, \beta_0 \ge 0$. The class $\mathcal{C}_{n,m}(s)$ of instances with these summaries is nonempty if and only if (3) holds together with

(i) $\beta = 0 \Rightarrow I_0 = \alpha a$, and (ii) $\beta_0 = \alpha \Rightarrow I_0 \ge \alpha a$.

Proof. Necessity: (3) is classical, and the two amplitude conditions are forced. Throughout, $v \in \mathbb{C}^n$ and $C \in \mathbb{C}^{n \times m}$, so $C^H : \mathbb{C}^n \to \mathbb{C}^m$ maps input space to output space. If $\beta = 0$, then $C^H P = 0$, so $C^H = (C^H v) v^H$ (equivalently, the range of C lies in span{v} in $\mathbb{C}^n$) and $C^H x = (C^H v)(v^H x)$ for every $x \in \mathbb{C}^n$; hence $I_0 = \| C^H v \| \, |v^H \hat{h}| = \alpha a$, and in particular $\alpha = \beta = \beta_0 = 0$ forces $I_0 = 0$. If $\beta_0 = \alpha > 0$, norm attainment at v gives $C C^H v = \alpha^2 v$, and splitting $\hat{h}$ along v and $v^\perp$ yields ${I_0}^2 = \alpha^2 a^2 + \| (I - \hat{d}\hat{d}^H) C^H \hat{h} \|^2 \ge \alpha^2 a^2$ with $\hat{d} = C^H v / \alpha$. Sufficiency: all instances below use $v = e_1$ and $b = \sqrt{p}\, e_1$, followed by zero padding to the fixed (n, m); the padded error coordinates enter neither the desired nor the interference term, so the infimum in (1) is unchanged. The branches are mutually exclusive and exhaustive: **(A)** zero operator ($\alpha = \beta = \beta_0 = 0$, hence $I_0 = 0$ by (i)): $C = 0$, $\hat{h} = a e_1$. **(B)** $\beta = 0 < \alpha$ (so $\beta_0 = \alpha$ and $I_0 = \alpha a$): $C^H = [\alpha, 0, \dots]$, $\hat{h} = a e_1$. **(C)** $\alpha = 0 < \beta$ (so $\beta_0 = \beta$): $C^H = [0, \beta]$, $\hat{h} = (a, I_0/\beta)^T$. **(D)** $0 < \alpha < \beta_0$: with $\delta = \sqrt{{\beta_0}^2 - \alpha^2} > 0$ and $\beta \ge \delta$ by (3), take $C^H = \begin{pmatrix} \alpha & \delta & 0 \\ 0 & 0 & \beta \end{pmatrix}$ and $\hat{h} = (a, (I_0 - \alpha a)/\delta, 0)^T$, for which $C^H \hat{h} = (I_0, 0)^T$, $\| C^H v \| = \alpha$, $\| C^H P \|_2 = \max\{\delta, \beta\} = \beta$ and $\| C \|_2 = \max\{\sqrt{\alpha^2 + \delta^2}, \beta\} = \beta_0$. **(E)** $\beta_0 = \alpha > 0$, $\beta > 0$ (so $I_0 \ge \alpha a$): $C^H = \mathrm{diag}(\alpha, \beta)$, $\hat{h} = (a, J_\perp/\beta)^T$ with $J_\perp = \sqrt{{I_0}^2 - \alpha^2 a^2}$, giving ${I_0}^2 = \alpha^2 a^2 + {J_\perp}^2$. Every branch realizes the six summaries exactly; (D) covers all $I_0 \ge 0$ with a single formula. □

2.3 Six-scalar information model and certification hierarchy

The six retained summaries define an information map, not a physical channel model. Let $\mathcal{I}(\mathcal{X})$ denote the six-scalar summary of a full configuration instance $\mathcal{X} = (b, \hat{h}, C)$ at a user, and let

$\mathcal{C}(s) = \{\mathcal{X} : \mathcal{I}(\mathcal{X}) = s\}$ (2c)

be the class of all instances indistinguishable to a certificate that observes only s. Throughout, the error radius r, the SINR level γ, the noise power σ^2 and the dimensions n, m are fixed external parameters of the certification problem; they are not part of the summary s and are never varied inside an information class. A six-scalar certificate is any function $B(s)$ of the retained summaries that must remain valid across the entire class, $B(s) \le q^*(\mathcal{X})$ for all $\mathcal{X} \in \mathcal{C}(s)$, and the strongest possible uniform certificate under this information budget is therefore

$B^*(s) = \inf_{\mathcal{X} \in \mathcal{C}(s)} q^*(\mathcal{X})$ (2d)

Sections 3–5 characterize this quantity from the retained spectral information: Section 3 solves the directional extremum forced by the operator origin of the summaries; Section 4 proves that the resulting envelope equals $B^*(s)$ on the nondegenerate class and characterizes the separate optimum of the boundary class $\beta_0 = \alpha$.

The information map defines a branched family of certification interfaces rather than a single linear ladder. With the same six scalars one may use the three gains separately (the separate-gain comparison envelope of this paper) or jointly through spectral compatibility (the envelope of this

paper). Two distinct one-scalar augmentations are of interest: adjoining $\kappa = \| A \hat{h} \|$, the norm of the linear coefficient of the quadratic objective, yields the quadratic-form bound $L_{quad} = c_q - 2r\kappa - \gamma {\beta_0}^2 r^2$ of Section 7; adjoining $H = \| \hat{h} \|$ restricts the class itself, for which we give a necessary compatibility condition but no complete envelope. Finally, the full quadratic data support cheap bounds and exact trust-region or S-procedure solvers. The two augmentations are different information branches and are not merged into a single seven-scalar class. Adding information can only leave the strongest uniform certificate B^* unchanged or tighter; it does not imply that any particular computable bound tightens, and indeed Q_G and L_{quad} exhibit no universal dominance relation in either direction. The interfaces also differ in cost structure: the six scalars and κ are built once from matrix geometry; their storage and transfer are then fixed-size; subsequent bound evaluation is scalar work; exact value queries, in contrast, require richer information, which can be stored as a dimension-dependent spectral cache without retaining the original matrices (Section 6.4).

Why not retain one more scalar? Nothing prevents it: this paper answers the optimality question for exactly one fixed interface, and richer summaries — κ, H, or others — are legitimate alternatives whose complete envelopes are open. The six-scalar interface is studied because, together with the nominal amplitudes required by the QoS envelope, it retains the three spectral quantities of one operator, so that joint spectral compatibility and the strongest uniform guarantee can be analyzed in closed form at a fixed summary size. We make no claim that six scalars are the best compression, that only this interface can be fixed-size, or any measured storage or communication saving.

Such a fixed-size interface can arise when candidate configurations are screened repeatedly, when configuration descriptors are cached or exchanged across network components, or when one wishes to evaluate a stored candidate at multiple uncertainty radii without retaining the complete quadratic geometry. These scenarios motivate the abstraction but are not assumptions required by the theory.

2.4 Physical interpretation of the retained summaries

The retained quantities separate the desired direction from its orthogonal complement and summarize how one interference operator acts on these two subspaces. In wireless terms: p is the desired-stream transmit power; a is the nominal desired-signal amplitude along the desired beam; I_0 is the nominal interference amplitude, whose square contributes to the interference power. The three spectral quantities describe how the interference operator maps channel error to interference: α is its response to error parallel to the desired beam; β is the largest amplification any error component orthogonal to the beam can suffer; β_0 is the largest amplification any error at all can suffer.

The three spectral quantities should not be viewed as independent uncertainty parameters. They are coupled observations of the same interference operator. Spectral compatibility is therefore a structural consistency condition: a large response in one direction restricts what the same operator can realize in orthogonal directions.

A bound that uses these gains separately may combine individually valid worst cases that cannot be realized simultaneously by one operator and one shared error vector. The proposed directional extremum removes exactly this incompatibility while retaining the same six scalar summaries. Its improvement over the separate-gain envelope therefore comes from using more of the structure already contained in the retained information, rather than from introducing additional channel information.

2.5 Related work

Robust transmission under imperfect CSI splits along the information available. Full-geometry worst-case design under norm-bounded errors is classical: S-procedure and semidefinite reformulations deliver exact or conservative constraints for MISO downlink, relay, secure, and RIS-aided systems, with [1] an early IRS-aided framework, [4] a current multi-functional RIS design under bounded cascaded CSI error, and [9] a representative worst-case SINR treatment that decomposes the steering-vector error into parallel and orthogonal components. Exact solution of the ball-constrained quadratic

rests on the trust-region and S-lemma literature [2],[3]. Decomposing an error into parallel and orthogonal components and bounding the two effects separately is a recurring device in this line; the comparison envelope of the present paper — the shared-budget bound $\min\{\beta_0 r, \alpha x + \beta y\}$ assembled from the two standard norm constraints — is our baseline in that spirit; we have not found this exact envelope stated as a method in the literature and do not attribute it to any specific paper. A second line compresses information probabilistically, using distributional summaries with conservative approximations. Our question differs from both: the uncertainty is deterministic and bounded, and the compressed object is a fixed set of spectral summaries of one operator.

Table 1. Positioning relative to the closest lines.

Line	Info	Retained data	Result type	Shared budget	Spectral compat.	Class-sharp	Boundary
Robust RIS/beamforming [1],[4],[9]	full	matrices	exact / conservative	yes (L1)	no	no	no
Trust region / S-lemma [2],[3]	full	matrices	exact	—	—	—	hard case
Probabilistic SINR bounds	distill.	moments	conservative	no	no	no	no
Buzano-type inequalities [5]	partial	norms	two-sided inequality	no	partial	no	no
PSD completion [6]	partial	specified entries	feasibility	—	—	—	—
Contraction completion [7]	partial	operator values	set description	—	partial	—	Parrott
Optimal recovery [8]	partial	finite data	worst-case radius	—	—	radius concept (related)	—
This work (six scalars)	partial	six scalars	safe + sharp envelope	yes	yes	yes (nondegenerate)	yes
This work (6+κ branch)	+κ	seven scalars	quadratic-form bound	yes	no	not claimed	no

The mathematics we need is classical but dispersed, and we use it at the formula level detailed in Appendix A. The cross-term bound in our extremum is a Buzano-type refinement of Cauchy–Schwarz [5]; the two-by-two positive-semidefinite completion behind our extremum is proved directly by a determinant computation, and the chordal completion theorem [6] and the contraction-completion framework of Parrott [7] place it in a wider theory without our result being a direct substitution of either; and the strongest-uniform-certificate quantity $B^*(s)$ is related to the information-based viewpoint of optimal recovery [8] — we do not claim equivalence with a radius of information in the sense of a recovery-error bound. To the best of our knowledge, within the searches reported in Appendix A, we have not found an existing result that evaluates the directional extremum in closed form under exactly the three retained spectral summaries and a shared error budget, nor an existing treatment of the β_0=α boundary's phase-coupled envelope; we do not claim these ingredients individually as new.

3 Spectral compatible directional envelope

Write $e = tv + w$, where $v^H w = 0$, $x = |t|$ and $\| w \| \leq y$. Put $d = C^H v$. For an orthogonal unit or subunit input u, put $\ell = C^H u$ and $s = \| \ell \| \leq \beta$. Since the map with columns v, u is a contraction, its compressed output Gram matrix is bounded by ${\beta_0}^2 I$. Define $\delta = \sqrt{{\beta_0}^2 - \alpha^2}$ and $\nu = \sqrt{{\beta_0}^2 - \beta^2}$. Cauchy–Schwarz and the determinant condition give

$$|d^H \ell| \leq \min\left\{\alpha s, \delta\sqrt{{\beta_0}^2 - s^2}\right\} \tag{4}$$

Consequently, define the directional envelope by the scalar extremum

$$R^2(x,y) = \max_{0\le s\le \beta}\left\{\alpha^2x^2 + s^2y^2 + 2xy\min\left\{\alpha s, \delta\sqrt{{\beta_0}^2 - s^2}\right\}\right\} \tag{5}$$

Theorem 1 — Directional envelope. Under (3), the extremum in (5) is given by

$$R(x,y) = \begin{cases} \sqrt{\alpha^2x^2 + \beta^2y^2 + 2\delta\nu xy} & \delta x \le \nu y \\ \beta_0\sqrt{x^2+y^2} & \nu y < \delta x < \alpha y \\ \alpha x + \delta y & \delta x \ge \alpha y \end{cases} \tag{6}$$

For every admissible error, $\| C^H e \| \le R(x,y)$, and

$$R(x,y) \le \min\left\{\beta_0\sqrt{x^2+y^2}, \alpha x + \beta y\right\} \tag{7}$$

Proof. Compatibility implies $\delta \le \beta$ and $\nu \le \alpha$. The two cross term bounds in (4) exchange order at $s = \delta$. On $[0,\delta]$, the objective is $(\alpha x + sy)^2$, whose maximum is at $s = \delta$. For $y > 0$, on $[\delta,\beta]$, change variables to $k = \sqrt{{\beta_0}^2 - s^2} \in [\nu,\alpha]$. The objective becomes

$$\alpha^2x^2 + {\beta_0}^2y^2 - y^2k^2 + 2xy\delta k \tag{8}$$

Its maximizing argument is $\delta x/y$ projected onto $[\nu,\alpha]$. The lower endpoint, interior and upper endpoint give the three branches in (6). Values agree on both branch boundaries. If $y = 0$, direct substitution into (5) gives $R = \alpha x$, so division by y is unnecessary. If $\alpha = 0$, compatibility forces $\beta_0 = \beta$ and $R = \beta y$. If $\beta = 0$, it forces $\beta_0 = \alpha$ and $R = \alpha x$. When ${\beta_0}^2 = \alpha^2 + \beta^2$, the middle region disappears and $R = \alpha x + \beta y$. The all zero operator is included by these limits. Inequality (7) follows from the norm cap and the linear cross term cap in (5). Finally, the compression and (4) bound every actual output, proving safety. □

For a fixed operator, R is a safe upper bound on the interference output, possibly strict. The next lemma shows that over the class of operators with the same three spectral summaries the bound is exact, so no smaller uniform interference bound follows from (α,β,β_0) alone. This is a statement about the direction problem only; the six-scalar QoS sharpness of Section 4 concerns the smaller class that also fixes a and I_0, where the nominal channel enters. The envelope is nondecreasing in either nonnegative input amplitude, as follows from (5); the same expression, together with the attainment construction of Lemma 2, yields the Lipschitz control used in Section 6: the envelope is a supremum of realized linear output norms, each Lipschitz with constants (α,β).

Lemma 2 (attainment over the three-spectral class). Fix compatible (α,β,β_0) and $x,y \ge 0$. Let $\mathcal{D}$ be the class of operators C on $n \ge 3$ inputs and $m \ge 3$ outputs with $\| C^H v \| = \alpha$, $\| C^H P \|_2 = \beta$, $\| C \|_2 = \beta_0$, let $u \perp v$ be unit, and let $e = xv + yu$. Then $\max_{C\in\mathcal{D}} \| C^H e \| = R(x,y)$.

Proof. Theorem 1 gives $\| C^H e \| \le R(x,y)$ for every $C \in \mathcal{D}$. For attainment, suppose first $\alpha > 0$. Let k be the projection of $\delta x/y$ onto $[\nu,\alpha]$ (with $k = \alpha$ if $y = 0$) and set $d = (\alpha,0,0)^T$, $\ell = \left(\delta k/\alpha, \beta_0\sqrt{1-k^2/\alpha^2}, 0\right)^T$. A direct computation gives $\| d \| = \alpha$, $d^H\ell = \delta k$, $\| \ell \|^2 = {\beta_0}^2 - k^2$, and the Gram matrix $[d,\ell]^H[d,\ell]$ has ${\beta_0}^2$ as an eigenvalue — its characteristic polynomial $(\alpha^2 - \lambda)\left({\beta_0}^2 - k^2 - \lambda\right) - \delta^2k^2$ vanishes at $\lambda = {\beta_0}^2$ because $\alpha^2 + \delta^2 = {\beta_0}^2$, and the other eigenvalue $\alpha^2 - k^2$ does not exceed it since $k \in [\nu,\alpha]$ — so $\| [d,\ell] \|_2{}^2 = {\beta_0}^2$ and $\| xd + y\ell \|^2 = \alpha^2x^2 + y^2\left({\beta_0}^2 - k^2\right) + 2xy\delta k = R^2(x,y)$ by the branch evaluation in the proof of Theorem 1. Let u' be a unit vector orthogonal to both v and u, and set $C^H = [d,\ell,\ell_2][v,u,u']^H$ with $\ell_2 = \beta e_3$; this is allowed since $\| \ell \| = \sqrt{{\beta_0}^2 - k^2} \le \beta$ (as $k \ge \nu$) and e_3 is orthogonal to d and ℓ. Then $C^Hv = d$ and $C^Hu = \ell$, so $\| C^Hv \| = \alpha$; C^HP acts as $\ell(u^Hz) + \ell_2(u'^Hz)$ on inputs $z \perp v$, so $\| C^HP \|_2 = \max\{\| \ell \|, \beta\} = \beta$; and $\| C \|_2{}^2 = \lambda_{\max}(CC^H)$ equals the largest eigenvalue of the block diagonal

Gram $\left[[\alpha^2, \delta k], [\delta k, \| \ell \|^2]\right] \oplus [\beta^2]$ — the other eigenvalue on the (d, ℓ) block is $\alpha^2 - k^2 \leq \alpha^2 \leq {\beta_0}^2$ since $k \in [\nu, \alpha]$ — so $\| C \|_2{}^2 = \max\{{\beta_0}^2, \beta^2\} = {\beta_0}^2$, i.e. $\| C \|_2 = \beta_0$. Finally $C^H e = xd + y\ell$ has norm $R(x, y)$. If $\alpha = 0$ (so $\beta_0 = \beta$), take $d = 0$, $\ell = (\beta, 0, 0)^T$, $\ell_2 = (0, \beta, 0)^T$ in the same construction: $\| C^H e \| = \beta y = R(x, y)$. If $\beta = 0$ (so $\beta_0 = \alpha$), take $C^H = \alpha z v^H$ with any unit $z \in \mathbb{C}^m$: then $\| C^H v \| = \alpha$, $C^H P = 0$ and $\| C \|_2 = \alpha$, and $\| C^H (xv + yu) \| = \alpha x = R(x, y)$ for every m, n; the zero operator is trivial. Zero padding reaches any $n \geq 3, m \geq 3$. □

4 QoS envelopes and sharpness

4.1 Safe shared budget bound

For $x = |v^H e|$, the reverse triangle inequality gives $|b^H(\hbar + e)| \geq \sqrt{p}|a - x|$, including $x > a$. The orthogonal budget is at most $\sqrt{r^2 - x^2}$. Define

$$Q_G(r) = \min_{0 \leq x \leq r} \left\{ p(a - x)^2 - \gamma \left[I_0 + R\left(x, \sqrt{r^2 - x^2}\right) \right]^2 - \gamma \sigma^2 \right\} \tag{9}$$

The separate-gain comparison bound $Q(r)$ replaces R in (9) with

$$R_{old}(x, y) = \min\{\beta_0 r, \alpha x + \beta y\}, \quad y = \sqrt{r^2 - x^2} \tag{10}$$

Proposition 1 — Safety and dominance. For every compatible fixed instance,

$$Q(r) \leq Q_G(r) \leq q^*(r) \tag{11}$$

Proof. For any ball feasible error, square the nonnegative lower bound $\sqrt{p}|a - x|$. Apply the interference triangle inequality and the directional envelope with the remaining orthogonal budget. This gives a pointwise lower bound by the objective in (9). Minimizing proves $Q_G \leq q^*$. Inequality (7), with $x^2 + y^2 = r^2$, makes the objective in (9) no smaller than the old objective, proving $Q \leq Q_G$. □

Thus, a positive safe numerical lower bound can accept a candidate. A nonpositive value returns unknown and requires a reference decision if a final classification is needed.

4.2 Nondegenerate information class sharpness

Theorem 2 — Sharpness away from the degenerate boundary. Fix $n \geq 5$, $m \geq 3$, $\beta > 0$, $\beta_0 > \alpha \geq 0$, $p > 0$, $\gamma > 0$, $r \geq 0$, $a, I_0 \geq 0$ and a summary vector s for which $\mathcal{C}_{n,m}(s)$ is nonempty (Lemma 1). With $r, \gamma, \sigma^2, n, m$ held fixed, the strongest uniform six-scalar certificate equals the spectral-compatible envelope:

$$\inf_{(b, \hbar, C) \in \mathcal{C}(s)} q^*(r) = Q_G(r) \tag{12}$$

The dimensions are sufficient for the construction; no minimal dimension claim is made. The theorem asserts three distinct facts: (a) Q_G is safe on the whole class (Proposition 1); (b) the class infimum of the true worst-case margin equals Q_G; and (c) an explicit construction sequence approaches the infimum. Finite attainment is proved in three cases: $r = 0$ (every instance attains it); $\alpha = 0$ with $\beta > 0$ (the explicit construction below attains (9) exactly); and the boundary class of Theorem 3 (attained by an explicit instance). For the remaining nondegenerate cases ($\alpha > 0, r > 0$), the construction does not furnish a uniform bound on the nominal-channel norm over the information class; divergence may occur in particular parameter families (Section 5 exhibits one), but is not asserted for every nondegenerate summary vector, and finite attainment is proved below for at least one nondegenerate instance (at the upper spectral endpoint). Whether every summary vector admits a finite-norm attaining instance is open.

Proof for positive alpha and radius. First take $\alpha > 0$ and $r > 0$. Continuity gives a minimizing $x \in [0, r]$ in (9); put $y = \sqrt{r^2 - x^2}$. If $y > 0$, let k be the projection of $\delta x / y$ onto $[\nu, \alpha]$. If $y = 0$, choose $k = \alpha$. In three output coordinates define

$$d = \begin{pmatrix} \alpha \\ 0 \\ 0 \end{pmatrix}, \quad \ell = \begin{pmatrix} \delta k/\alpha \\ \beta_0\sqrt{1-k^2/\alpha^2} \\ 0 \end{pmatrix} \tag{13}$$

These vectors satisfy $\| d \| = \alpha$, $\| \ell \| \leq \beta$, $d^H \ell = \delta k$ and $\| xd + y\ell \| = R(x,y)$. For $Z = [d, \ell]$, the two by two matrix $Z^H Z$ has largest eigenvalue ${\beta_0}^2$: the determinant of ${\beta_0}^2 I - Z^H Z$ is zero, and its diagonal entries are nonnegative — a direct two-by-two determinant computation, not an invocation of a general completion theorem. Therefore,

$$dd^H + \ell\ell^H \preccurlyeq {\beta_0}^2 I_3 \tag{14}$$

Choose $0 < \varepsilon < 1$ and put $\ell_\varepsilon = (1-\varepsilon)\ell$. In output space, rank one scaling gives the valid PSD ordering

$$dd^H + \ell_\varepsilon {\ell_\varepsilon}^H \preccurlyeq dd^H + \ell\ell^H \preccurlyeq {\beta_0}^2 I_3 \tag{15}$$

No ordering between the original and contracted two by two Gram matrices is required. In fact, their difference generally has a negative determinant when the cross term is nonzero.

Let $d_2, \ell_{\varepsilon,2}$ be the first two coordinates, and define a planar interpolant

$$S(t) = d_2 {d_2}^H + (1-t)\ell_{\varepsilon,2}{\ell_{\varepsilon,2}}^H + t\beta^2 I_2 \tag{16}$$

We claim that $\lambda_{\max}\big(S(0)\big) < {\beta_0}^2$. Otherwise a unit vector attaining ${\beta_0}^2$ for $S(0)$ also attains equality in (14). Since the removed PSD term is $(2\varepsilon - \varepsilon^2)\ell_2 {\ell_2}^H$, that vector is orthogonal to ℓ_2. Its eigenvalue for $S(0)$ must then come entirely from $d_2 {d_2}^H$, which is at most $\alpha^2 < {\beta_0}^2$, a contradiction. At the other endpoint, $\lambda_{\max}\big(S(1)\big) = \alpha^2 + \beta^2 \geq {\beta_0}^2$. Continuity therefore gives a $t_\varepsilon \in (0,1]$ for which $\lambda_{\max}\big(S(t_\varepsilon)\big) = {\beta_0}^2$. An explicit root or uniqueness is not needed. Using the planar eigenvalue is essential when $\beta = \beta_0$: an auxiliary third direction could otherwise make the full maximum eigenvalue constant over an initial interval of t.

Complete the orthogonal operator through

$$T_\varepsilon = \operatorname{diag}\Big((1-t_\varepsilon)\ell_{\varepsilon,2}{\ell_{\varepsilon,2}}^H + t_\varepsilon\beta^2 I_2, \beta^2\Big) \tag{17}$$

The planar block is bounded by $\beta^2 I_2$, and the third direction equals β^2. Hence $\| T_\varepsilon \|_2 = \beta^2$ and $\| dd^H + T_\varepsilon \|_2 = {\beta_0}^2$. Moreover,

$$M_\varepsilon = T_\varepsilon - \ell_\varepsilon {\ell_\varepsilon}^H = \operatorname{diag}\big(t_\varepsilon\big(\beta^2 I_2 - \ell_{\varepsilon,2}{\ell_{\varepsilon,2}}^H\big), \beta^2\big) \succ 0 \tag{18}$$

Positive definiteness follows from $t_\varepsilon > 0$, $\beta > 0$ and $\| \ell_\varepsilon \| < \beta$; $K_\varepsilon = {M_\varepsilon}^{1/2}$ is invertible, so $B_\varepsilon = [\ell_\varepsilon, K_\varepsilon]$ has full row rank three. Set

$$B_\varepsilon = [\ell_\varepsilon, K_\varepsilon], \quad {C_\varepsilon}^H = [d, B_\varepsilon], \tag{19}$$

with three rows and five columns. With $v = e_1$ and $b = \sqrt{p}\, e_1$, we obtain $B_\varepsilon {B_\varepsilon}^H = T_\varepsilon$ and the exact values α, β, β_0; the nominal channel below fixes the remaining three summaries exactly.

Set $u_\varepsilon = -xd - y\ell_\varepsilon$. Its norm is nonzero: $\delta > 0$, the chosen ℓ is nonzero, its first coordinate is nonnegative, and x, y are not both zero. Define

$$z_\varepsilon = I_0 \frac{u_\varepsilon}{\|u_\varepsilon\|}, \quad B_\varepsilon \hat{h}_\perp = z_\varepsilon - ad, \quad \hat{h}_\varepsilon = (a, \hat{h}_\perp)^T \tag{20}$$

Full row rank ensures that this linear equation has a solution, and ${C_\varepsilon}^H \hat{h}_\varepsilon = z_\varepsilon$, so a and I_0 are matched exactly, including $I_0 = 0$. For $e_\varepsilon = (-x, -y, 0, 0, 0)^T$, the ball norm equals r and ${C_\varepsilon}^H e_\varepsilon = u_\varepsilon$: nominal and perturbation interference are aligned, and the signal phase is worst-case simultaneously — the error components $-x$ and $-y$ are jointly feasible and share the phase that realizes the envelope. Consequently,

$$q_\varepsilon(e_\varepsilon) = p(a-x)^2 - \gamma(I_0 + \| u_\varepsilon \|)^2 - \gamma\sigma^2 \tag{21}$$

Let $R = R(x, y)$ and $R_\varepsilon = \| u_\varepsilon \|$. Triangle inequalities imply $0 \le R - R_\varepsilon \le \varepsilon y \, \| \ell \| \le \varepsilon y \beta$, because the constructed vectors have nonnegative real coordinates; this is where the nonnegativity of the coordinates of d and ℓ_ε is used. Therefore,

$$0 \le q_\varepsilon(e_\varepsilon) - Q_G(r) \le \gamma(2I_0 + 2R)\varepsilon y \beta \tag{22}$$

Since $Q_G \le {q_\varepsilon}^* \le q_\varepsilon(e_\varepsilon)$ and $\varepsilon > 0$ is arbitrary, the class infimum is at most Q_G. Proposition 1 proves the reverse inequality, giving (12).

Remark (bounded construction at the upper spectral endpoint). At $\beta_0 = \sqrt{\alpha^2 + \beta^2}$ we have $\delta = \beta$ and $\nu = \alpha$, so the projection gives $k = \alpha$ for every (x, y): $d = \alpha e_1$ and $\ell = \delta e_1$ are collinear, $t_\varepsilon = 1$, and $u_\varepsilon = -(\alpha x + (1 - \varepsilon)\delta y)e_1$ points along e_1 for every ε. The right-hand side of (20) then lies in the span of e_1, and the minimum-norm solution has ε-independent norm; the construction family therefore keeps $\| \hat{h}_\varepsilon \|$ bounded. Numerically, for $p = 2$, $a = I_0 = \alpha = \beta = \gamma = 1$, $\beta_0 = \sqrt{2}$, $\sigma^2 = 0.05$, $r = 0.2$ (minimizer $x^* \approx 0.18332$, $y^* \approx 0.07995$), $\| \hat{h}_\varepsilon \| = \sqrt{5}$ for every tested ε, while $q_\varepsilon(e_\varepsilon) - Q_G = O(\varepsilon)$ as guaranteed by (22). At this endpoint the infimum is in fact attained finitely: with $C^H = \big[[1,1,0,0,0], [0,0,0,1,0], [0,0,0,0,1]\big]$, $b = \sqrt{2}e_1$ and $\hat{h} = (1, -2, 0, 0, 0)^T$, the six summaries are exactly $\big(2,1,1,1,1,\sqrt{2}\big)$ and the error $e = (-x^*, -y^*, 0, 0, 0)^T$ satisfies $C^H e = -(x^* + y^*)e_1$ parallel to $C^H \hat{h} = -e_1$, so $q(e) = 2(1 - x^*)^2 - (1 + x^* + y^*)^2 - 0.05 = Q_G = q^*$ (verified to 7×10^{-16}). We do not claim finite attainment for the whole upper-endpoint family.

Zero alpha and zero radius. If $\alpha = 0$, compatibility gives $\beta_0 = \beta$. Use $C^H = [0, \beta]$, $\hat{h} = (a, I_0/\beta)^T$ and $b = \sqrt{p}\, e_1$. For $e = (-x, y)^T$, the signal equals $p(a - x)^2$ and interference equals $(I_0 + \beta y)^2$. Since $R = \beta y$, choosing a minimizing x attains (9) exactly, so here the infimum is attained. If $r = 0$, every compatible instance has the same margin $pa^2 - \gamma {I_0}^2 - \gamma\sigma^2 = Q_G(0)$, and the infimum is attained trivially.

Padding to general dimensions. Given any of the above instances with $n' \le n$ inputs and $m' \le m$ outputs, append $n - n'$ zero coordinates to every input vector (b, $\hat{h}$, the columns of C, and the error) and $m - m'$ zero rows to C^H. All six summaries, the error norm, the ball feasibility and every objective value are unchanged — the padded error coordinates enter neither the desired nor the interference term, so the infimum in (1) is unchanged — and the construction transfers verbatim to any fixed $n \ge 5, m \ge 3$. The proof covers the upper spectral endpoint $\beta_0 = \sqrt{\alpha^2 + \beta^2}$ and the case $\beta = \beta_0$ without excluding them. □

4.3 Degenerate boundary and explicit attainment

Suppose $\beta_0 = \alpha > 0$. The equality $\beta_0 = \alpha$ is itself one of the six summaries; what changes at the boundary is the structure it forces. Norm attainment at v gives $CC^H v = \alpha^2 v$, so the interference generated by any orthogonal error component is orthogonal to the interference generated along the beam: the parallel interference output is a fixed complex multiple of the desired signal output, and the two share one phase within the whole boundary class. In nondegenerate classes the phases can be chosen independently by suitable instances of the class — a statement about the class, not about any single fixed operator. The completion construction of Theorem 2 exploits precisely this class-level freedom, which is why the boundary needs a separate analysis. Put $s = v^H \hat{h}$, $\hat{d} = d/\alpha$ and

$$J_\perp = \sqrt{{I_0}^2 - \alpha^2 a^2}, \quad c_\parallel = p - \gamma\alpha^2 \tag{23}$$

Boundary compatibility is the nonemptiness condition of Lemma 1: $I_0 \ge \alpha a$ and $0 \le \beta \le \alpha$; if $\beta = 0$, then $I_0 = \alpha a$ and $J_\perp = 0$. The margin decomposes exactly as

$$q(e) = c_\parallel |s + t|^2 - \gamma \, \| (I - \hat{d}\hat{d}^H) C^H (\hat{h} + w) \|^2 - \gamma\sigma^2 \tag{24}$$

Theorem 3 — Boundary sharpness. On a nonempty boundary information class with $\beta_0 = \alpha > 0$ and $\beta > 0$, for fixed dimensions $n, m \ge 2$, the minimum information class margin is

$$Q_\partial(r) = \min_{0\le x\le r}\left\{c_\parallel F_{c_\parallel}(a,x) - \gamma\left[J_\perp + \beta\sqrt{r^2-x^2}\right]^2 - \gamma\sigma^2\right\}, \tag{25}$$

$$F_{c_\parallel}(a,x) = \begin{cases}(a-x)^2 & c_\parallel \ge 0\\ (a+x)^2 & c_\parallel < 0\end{cases} \tag{26}$$

In particular, $\min_{\mathcal{C}(s)} q^*(r) = Q_\partial(r)$, and this value is attained by an explicit instance.

Proof of safety. In (24), if $c_\parallel \ge 0$, the reverse triangle inequality gives $c_\parallel|s+t|^2 \ge c_\parallel|a-x|^2 = c_\parallel(a-x)^2$. If $c_\parallel < 0$, then $|s+t| \le a+x$, so multiplying the inequality by the negative number $c_\parallel$ reverses it: $c_\parallel|s+t|^2 \ge c_\parallel(a+x)^2$. The nominal orthogonal interference has norm $J_\perp$; its perturbation has norm at most βy, with $y \le \sqrt{r^2-x^2}$. The triangle inequality and minimization yield $Q_\partial \le q^*$.

Explicit construction. For $\beta > 0$, take

$$C^H = \begin{pmatrix}\alpha & 0\\ 0 & \beta\end{pmatrix},\ \hbar = \begin{pmatrix}a\\ J_\perp/\beta\end{pmatrix},\ b = \begin{pmatrix}\sqrt{p}\\ 0\end{pmatrix} \tag{27}$$

The six summaries are exactly $p, a, I_0, \alpha, \beta, \alpha$, since $\beta \le \alpha$ and $\alpha^2 a^2 + {J_\perp}^2 = {I_0}^2$. Let x^* minimize (25) and put $y^* = \sqrt{r^2 - x^{*2}}$; choose $e = (-x^*, y^*)^T$ for $c_\parallel \ge 0$, or $e = (x^*, y^*)^T$ for $c_\parallel < 0$. At this common point, the parallel term in (24) equals $c_\parallel F_{c_\parallel}(a, x^*)$ and the orthogonal interference norm equals $J_\perp + \beta y^*$: the signal phase and the orthogonal error are optimized jointly at the same minimizing x^*. The error is feasible and its objective equals Q_∂; safety gives equality with the class minimum at this instance. Zero padding establishes attainment in larger dimensions. If $\beta = 0$ (so $I_0 = \alpha a$), take $C^H = [\alpha, 0, \dots]$, $\hbar = a e_1$ and $b = \sqrt{p}\, e_1$; the same phase choice attains the parallel formula, which is (25) with $J_\perp = 0$. At $r = 0$, no limiting argument is needed. □

The general safe bound Q_G remains valid on the boundary, while Q_∂ incorporates the phase compatibility that the boundary class enforces. Comparing them for the same boundary summaries is legitimate. Applying Q_∂ to an instance with $\beta_0 > \alpha$ is not justified.

5 Boundary behavior and additional channel norm information

Take $p = 2$, $a = I_0 = \alpha = \gamma = 1$, $\beta = 0.7$, $\sigma^2 = 0.05$ and $r = 0.2$. The boundary construction (27) has $J_\perp = 0$, $c_\parallel = 1$ and $\hbar = (1,0)^T$. Its one dimensional objective is $(1-x)^2 - 0.49(0.04 - x^2) - 0.05$, which decreases on $[0,0.2]$. Thus $Q_\partial = q^* = 0.59$, attained at $x = 0.2$. The general safe bound for these boundary summaries is $Q_G = -0.21$.

For comparison, consider a nondegenerate family with $C^H = \begin{pmatrix}1 & \delta & 0\\ 0 & 0 & 0.7\end{pmatrix}$, $\hbar = (1, -2/\delta, 0)^T$ and $b = \left(\sqrt{2}, 0, 0\right)^T$. It preserves the other five summaries while $\beta_0 = \sqrt{1+\delta^2}$ approaches α from above. For $\delta = 0.1, 0.03, 0.01$, the reference and Q_G agree to numerical accuracy at −0.211028, −0.210093 and −0.210010. The nominal channel norms are 20.02, 66.67 and 200.00. The objective at $e = (-0.2, 0, 0)^T$ is always −0.21. The nominal channels do not approach a finite fixed channel: when nondegenerate families approach the boundary, the class over which the infimum is taken changes, so the optimum jumps — an information-class discontinuity, not a physical one.

Table 2. Explicit boundary checks with a = 1 throughout.

p	a	I₀	J⊥	c‖	Q∂	Reference q*
2	1	1	0	1	0.590000	0.590000
0.2	1	1	0	−0.8	−1.202000	−1.202000
2	1	1.3	0.830662	1	−0.181020	−0.181020
0.2	1	1.3	0.830662	−0.8	−1.963482	−1.963482

If the channel norm $H = \| \hat{h} \|$ is additionally retained, set $z = C^H \hat{h}$. A necessary phase compatibility condition is

$$|d^H z - \alpha^2 s| \le \alpha \sqrt{{\beta_0}^2 - \alpha^2} \sqrt{H^2 - a^2} \tag{28}$$

Indeed, $(CC^H - \alpha^2 I)v$ is orthogonal to v and its squared norm is at most $\alpha^2\left({\beta_0}^2 - \alpha^2\right)$. Cauchy–Schwarz with the orthogonal nominal channel component proves (28). This condition excludes some constructions requiring an unbounded channel norm; the complete envelope of the H-restricted class is left open. The H-branch and the κ-branch of Section 2.3 are different augmented interfaces and are not merged.

6 Numerical certification and reference interface

6.1 Interval lower bounds

A sampled minimum is an upper estimate of a continuous minimum and must not be used for acceptance. We set $x = r\sin\theta$, $y = r\cos\theta$ and partition $[0, \pi/2]$ into N_θ cells. For a cell midpoint θ_m and width $\Delta\theta$, use

$$R_U = \min\{\beta_0 r, R(r\sin\theta_m, r\cos\theta_m) + r(\alpha + \beta)\Delta\theta/2\} \tag{29}$$

The Lipschitz bound $|\Delta R| \le \alpha|\Delta x| + \beta|\Delta y|$ follows by representing the direction envelope as the supremum of the compatible linear output norms; Lemma 2 attains the scalar extremum for each direction, including degenerate spectra. Each fixed norm is Lipschitz with these constants, so their supremum is as well.

For the signal, use p times the squared distance from a to $[r\sin\theta_{lo}, r\sin\theta_{hi}]$. Subtract $\gamma(I_0 + R_U)^2 + \gamma\sigma^2$ and take the minimum cell lower bound. This produces $Q_{G,LB} \le Q_G$ in real arithmetic. The separate-gain envelope uses the same partition and cap. Main comparisons use $N_\theta = 256$; refinements to 1024 and 4096 are reported separately. The analytical inequality is exact; the reported implementation uses floating-point arithmetic with conservative numerical safeguards rather than formally verified directed rounding.

6.2 Engineering threshold and reference solver

For reference evaluation define $A = bb^H - \gamma CC^H$, $g = A\hat{h}$ and $c_q = \hat{h}^H A\hat{h} - \gamma\sigma^2 = pa^2 - \gamma {I_0}^2 - \gamma\sigma^2$. The margin is $c_q + e^H Ae + 2\mathrm{Re}(e^H g)$. The common engineering threshold is

$$\tau = 10^{-8}\max\{1, |c_q|, 2r \| g \|, r^2 \| A \|_2\} \tag{30}$$

The theoretical envelope depends only on six summaries; this experimental threshold and the reference validity checks additionally use full quadratic data. A method directly accepts a candidate only if every user's implemented lower bound exceeds its threshold. Otherwise it returns unknown. Reference statuses are clearly feasible for $q^* > \tau$, clearly infeasible for $q^* < -\tau$, ambiguous near zero, and invalid if numerical validity checks fail. Invalid references remain explicitly counted and cannot establish a rejection or a safety conclusion.

The reference solves the Hermitian trust region problem by eigenvalue decomposition and a multiplier search, handling PSD interior solutions, singular cases and the indefinite hard case. The global optimality conditions are stationarity, ball feasibility, $A + \mu I \succcurlyeq 0$, $\mu \ge 0$ and complementary slackness [2],[3]. Numerical checks inspect the full stationarity residual, shifted minimum eigenvalue, ball violation and complementarity. A 106 case audit reports a maximum primal–dual gap of 4.97×10^{-14}. The fixed 200 iteration multiplier bisection used in early timing over resolves the multiplier: stopping the bisection once the bracket width falls below 10^{-12} of its magnitude changes no computed value by more than 5.7×10^{-10}, preserves every validity check, and uses a median of 40 iterations. A 10^{-10} stop already shifts some values by up to 2.8×10^{-8} and is not used. All pipeline

timings in Section 7 use the tolerance-stopped reference; a supplementary table reproduces the original 200 iteration protocol for transparency.

Fallback skips users already certified by the lower bound and stops at the first clearly infeasible user. Each user's eigenvalue factorization is reused across radii. Screening and reference calculations do not read the truth pass cache during timing. These implementation choices affect measured runtime and are not theoretical contributions.

6.3 S procedure SDP baseline

As a full information baseline independent of our eigendecomposition reference, we adapt the standard single sphere S procedure LMI to the fixed candidate certification problem: for each user, maximize t subject to $\lambda \geq 0$ and

$$\begin{pmatrix} A + \lambda I & g \\ g^H & c_q - t - \lambda r^2 \end{pmatrix} \succcurlyeq 0, \tag{31}$$

whose optimum equals $q^*(r)$ by the S lemma; the LMI is solved on the real embedding of the Hermitian data with an interior point solver [2],[3],[4]. This is a classical technique, representative of its use in robust RIS design [4], adapted to our certification interface; it is not a reproduction of that literature's system level algorithms and is not claimed as a new method.

The SDP provides a lower bound, but a negative lower bound proves nothing about infeasibility, and a solver status of optimal is not a certificate. Our decision rule is two sided. The lower bound L is the residual corrected dual feasible value: the shift $(t, \lambda) \to (t - \eta(1 + r^2), \lambda + \eta)$ adds blockdiag$(\eta I, \eta)$ to the LMI, so Weyl's inequality restores positive semidefiniteness up to eigensolver rounding, with an empirical rounding allowance and a numerical recheck; this is an implementation detail, not a formal error bound. Infeasibility is certified only by an upper bound: from the corrected multiplier we recover the Lagrangian minimizer $(A + \lambda' I)e = -g$, project it onto the ball if $\| e \| > r$, and re evaluate the true objective $U = q(e_w)$. Since e_w is ball feasible, U is a valid upper bound of q^*, and $U < -\tau$ certifies infeasibility; no optimality of e_w is claimed. All remaining cases return unresolved and are never treated as rejections.

6.4 Compact spectral value queries and guarded decisions

For fixed γ and σ^2, write $A = U\, diag(\lambda) U^H$, $g = A\,\hat{h}$, $w_i = |(U^H g)_i|^2$ and $c_q = \hat{h}^H A \hat{h} - \gamma\sigma^2$. The cache (λ, w, c_q) supports repeated optimal-value queries without the original matrices or eigenvectors. A scale field is retained for numerical stopping; at dimension eight the float64 payload is 144 bytes. Original-coordinate error-witness recovery is a different task and may require U.

For $r > 0$ and $\mu \geq \max\{0, -\lambda_{\min}\}$ with $\lambda_i + \mu > 0$ whenever $w_i > 0$, the dual value is $d(\mu) = c_q - \mu r^2 - \Sigma_i\, w_i/(\lambda_i + \mu)$; it is a lower bound on the optimum. Exactly zero weights contribute zero, including at singular denominators. The squared norm of the stationary spectral vector is $\Sigma_i\, w_i/(\lambda_i + \mu)^2$. If this norm is at most r^2 at $\mu = 0$ for a positive semidefinite operator, the interior value is attained. At a negative-curvature endpoint with exactly zero weight on the minimum eigenspace, the unused norm is filled in that eigenspace and the endpoint dual value is attained. Otherwise, bracketed bisection maintains a feasible upper multiplier. The stationary vector at this multiplier supplies a feasible primal upper value, so the dual–primal gap provides an objective-based stopping rule. No nonzero weight is discarded to force a hard case.

The implementation retains lower and upper values with a scale-dependent floating-point guard and stops at relative objective gap 10^{-12} or at multiplier representability limits. Acceptance requires the lower value to exceed the common threshold; rejection requires the upper value to be below its negative; otherwise the query is unresolved. These guards are numerical diagnostics, not outward-rounded interval proofs. Eigenvalue and cache construction are also floating-point operations. The compact cache is an additional implementation of the same classical trust-region problem, not a theoretical contribution.

A regression case explains why small nonzero weights must be preserved: in dimension eight take $b = e_2$, $C = [e_1, 0,0]$, $\hat{n} = -10^{-6}e_1 + e_2$, $\gamma = 1$, $\sigma^2 = 0.1399999$ and $r = 0.6$. The feasible error $e = -\sqrt{0.11}e_1 - 0.5e_2$ has objective -5.6333×10^{-7}. Treating its minimum-eigenspace weight as zero can incorrectly yield a positive value near 10^{-7}. The repaired implementation rejects this case and checks both zero and small nonzero weights. Across 864 pool queries at three radii and three SINR levels, its maximum difference from the tolerance-stopped reference is 1.94×10^{-12}; analytic and near-hard-case regressions are recorded separately. Agreement on these cases does not establish formal numerical certification for all inputs.

7 Numerical verification

7.1 Analytical implementation checks

With seed 20261006, 30 random compatible spectral tuples are checked against a 4097 point scalar grid, local refinement and direct evaluation of $s = 0, \delta, \beta$. The maximum $|R^2 - {R_{ref}}^2|$ is 1.11×10^{-16}. The grid only underestimate reaches 4.12×10^{-5} and is reported separately. Random inequality checks show no violation of (7). The three fixed direction cases cover the three branches; hand checks include zero spectral gains and zero input amplitudes.

The direction completion matches the three spectral summaries to machine precision (nine formal cases, and 251 additional repaired constructions covering asymmetric tuples, the upper spectral endpoint, $\beta = \beta_0$ and direction endpoints; maximum six scalar discrepancy 2.89×10^{-15}, objective identity residual 1.25×10^{-14}). The nominal channel construction matches all six summaries, and the deficit $q(e_\varepsilon) - Q_G$ obeys the proven bound (22) and falls by roughly one decade per decade of ε. A 36 case degenerate battery covers $\alpha = 0$, $\beta = 0$, $r = 0$, $I_0 = 0$, both spectral endpoints, both branch connection points, the trust region hard case equality, repeated minimum eigenvalues and radius equality. For the deterministic flip, the separable bound is −0.0425, the separate-gain approximate minimum is −0.024580 and the new approximate minimum is 0.034624; their interval lower bounds at 256 cells are −0.027068 and +0.027262, and the new lower bound at 65536 cells is 0.0345956. The positive certified lower bound, rather than the approximate minimum, establishes acceptance. These are consistency checks of the analytical proofs, not substitutes for them.

7.2 Controlled threshold-crossing examples

The scan fixes the flip example's other summaries and varies β_0 over 21 equally spaced points from 1.02 to $\sqrt{2}$. Each point has an explicit matrix and nominal channel; the channel norm ranges from 2.22 to 9.05. This is an instance family, not an experiment holding a physical channel fixed. The new envelope exceeds the separate-gain envelope at 20 of 21 points, with maximum gain 0.0749 at $\beta_0 = 1.1777$; they agree at $\beta_0 = \sqrt{2}$, as predicted by the spectral endpoint. Six points have $Q_{G,LB} > \tau$ while the separate-gain lower bound is nonpositive, with $\beta_0 \in [1.0791, 1.1777]$. The minimum reference to new interval lower bound gap is 0.00663325.

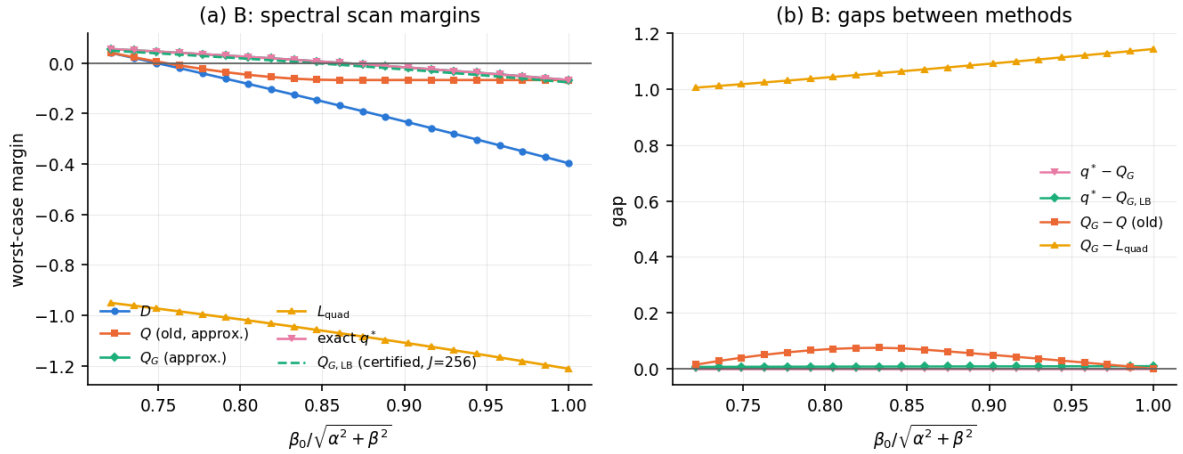


Figure 1. Constructed spectral scan. Approximate envelopes are diagnostics; the dashed interval lower bound is used for certification. The family changes the nominal channel norm.

7.3 Fixed-candidate certification study

The candidate model has two base stations with four antennas each, four users, and a 16 element RIS with two bit phases. Each seed generates six phase states (all ones plus five random) and two beam types, MRT and RZF (regularization 0.1), giving 12 candidates; per base station power is 1 with 0.5 per stream. The effective link is $\hbar_{j,u} = s_{j,u}\left[d_{j,u} + G_j{}^H diag(\theta) f_u\right]$ with same-cell scale 1 and cross-cell scale 0.5; entries of $G_j, d_{j,u}$ have circular complex normal variance 1/4 and entries of f_u variance 1/16. Channels and beams are stacked into dimension eight; noise power is 0.05. These are normalized dimensionless links, not a fitted propagation model.

The original pool (seeds 11–55, $\gamma = 0.5$, $\rho \in \{0.01, 0.03, 0.06\}$, 720 user problems, 180 units) is an easy baseline: all four methods accept all 180 units, with no false acceptance, no fallback, and zero additional accepts for the new envelope over the separate-gain envelope. The exploratory grid adds seeds 66–110 and evaluates $\gamma \in \{0.5, 1, 2\}$ and $\rho \in \{0.06, 0.15, 0.30\}$ — 1080 units and 4320 user problems, every cell reported, no selection by the new bound's gain. All references are valid; 472 units are clearly feasible and 608 clearly infeasible. There is no observed false acceptance or lower bound violation. At 256 cells the new bound directly accepts 390 units versus 380 for the separate-gain envelope (feasible coverage 82.63% versus 80.51%; original seeds contribute six additional accepts, added seeds four).

Table 3. Directly accepted units on the complete sensitivity grid, 256 cells.

γ	ρ	Feas.	Infeas.	D	Old Q	Q_G	L_quad	Added
0.5	0.06	118	2	117	118	118	118	0
0.5	0.15	108	12	91	102	103	102	1
0.5	0.30	63	57	6	26	28	21	2
1	0.06	90	30	78	81	83	90	2
1	0.15	56	64	21	34	36	49	2
1	0.30	7	113	0	0	0	0	0
2	0.06	24	96	13	18	18	22	0
2	0.15	6	114	0	1	4	6	3
2	0.30	0	120	0	0	0	0	0
Total		472	608	326	380	390	408	10

The quadratic-form bound with one additional gradient-norm summary uses

$$L_{quad}(r) = c_q - 2r\kappa - \gamma\beta_0{}^2 r^2, \;\; \kappa = \| A\,\hat{n} \| \tag{32}$$

Its construction computes κ from the matrix geometry, but its storage and evaluation need only the six summaries plus κ — the $6 + \kappa$ branch of Section 2.3; it is not a full-matrix method at evaluation time. In the grid it exceeds the implemented new interval bound on 2903 of 4320 user evaluations; that count does not compare it with the exact continuous envelope. A separate witness has $L_{quad} = 0.198195750531$ and a new envelope objective value of 0.196451952840 at a feasible one dimensional argument, so the continuous minimum is strictly below L_{quad} there; conversely, the constructed spectral family has $Q_G > L_{quad}$. The bounds therefore have no universal dominance relation; their maximum is safe and yields the highest direct-certification coverage among tested bounds. The pools overlap on 240 user problems; their combined 5040 evaluation records represent 4800 distinct user problems, and multiple radii or thresholds on one candidate are correlated, not independent samples.

7.4 Independent validation

Before computing any result on new seeds, we wrote a fixed configuration file recording the unused seeds 211–215, the grid, the candidate rules, the acceptance rule, the reference validity criteria, the timing protocol and SHA-256 hashes of the code files. The batch adds 540 units and 2160 user problems with zero overlap; the union of all pools contains 6960 distinct user problems. All references are valid and no false acceptance or bound violation occurs. Of 540 units, 159 are clearly feasible; direct accepts are D 111, separate-gain 125, new 126 and L_{quad} 138. The new bound adds one unit over the separate-gain envelope (coverage 78.62% to 79.24%).

Table 4. Preregistered validation batch (seeds 211–215, 540 units, 256 cells).

γ	ρ	Feasible	D	Old Q	Q_G	L_quad
0.5	0.06	54	53	53	53	54
0.5	0.15	44	28	36	37	40
0.5	0.30	13	0	4	4	5
1	0.06	29	22	23	23	28
1	0.15	15	5	6	6	8
1	0.30	0	0	0	0	0
2	0.06	4	3	3	3	3
2	0.15	0	0	0	0	0
2	0.30	0	0	0	0	0
Total		159	111	125	126	138

Table 5 separates interval resolution effects: in the exploratory grid the separate-gain envelope gains one accept at 4096 cells while the new envelope is unchanged; in the validation batch the counts are unchanged at every resolution. Decomposing the gap on the 2160 validation rows, the numerical interval loss (upper estimate minus the 256 cell certified bound) has median 1.22×10^{-3} and maximum 2.53×10^{-2}, whereas the theoretical envelope gap (reference minus upper estimate) has median 6.08×10^{-2} and maximum 2.68: certification outcomes are limited by the theoretical envelope rather than by the interval discretization.

Table 5. Interval resolution ablation (accepted units).

Cohort	Cells	Old Q	Q_G	Added	Feasible
Exploratory	256	380	390	10	472
Exploratory	1024	380	390	10	472

Cohort	Cells	Old Q	Q_G	Added	Feasible
Exploratory	4096	381	390	9	472
Validation	256	125	126	1	159
Validation	1024	125	126	1	159
Validation	4096	125	126	1	159

A second validation flavor uses an illustrative heterogeneous-link pool (seeds 301–305, same 540-unit structure): two base stations and a 16 element RIS with fixed geometry; user links with distance-dependent path loss $PL(d) = 10 + 30log_{10}(d/10)$ dB, a fixed 12 dB BS–RIS link, Rician factor $K = 3$ on BS–RIS and RIS–UE links and Rayleigh BS–UE links; the same candidate rules; per base station power 1 with 0.5 per stream; noise power 0.05; relative balls $r = \rho \parallel \hat{h} \parallel$. The radii are interpreted through the deterministic energy-ratio reading $\parallel e \parallel^2 / \parallel \hat{h} \parallel^2 \leq \rho^2$, whose dB value is $20log_{10}\rho$ (−24.44, −16.48, −10.46 dB for the three radii); this is a worst-case ball interpretation, not a statistical NMSE, which would require a separately specified error distribution, normalization and coverage probability. The pool is illustrative heterogeneous scaling, not a calibrated system model. All references are valid, no bound violation occurs, and the main conclusions reappear: of 540 units (211 feasible), the separate-gain envelope accepts 156, the new envelope 163 (77.25% versus 73.93% of feasible; seven units flipped), L_{quad} 186, and the joint maximum 190 (90.05%).

Table 6. Illustrative heterogeneous-link pool (seeds 301–305; distance-dependent path loss, Rician K=3 and Rayleigh components; 540 units, 256 cells). r = ρ‖ĥ‖; 20log10(ρ) is a deterministic worst-case energy-ratio reading, not a statistical NMSE.

γ	ρ	Feasible	Old Q	Q_G	L_quad	Joint	Flips
0.5	0.06	59	57	59	59	59	2
0.5	0.15	49	40	41	44	45	1
0.5	0.30	23	6	7	11	13	1
1	0.06	39	29	31	38	38	2
1	0.15	22	11	11	19	19	0
1	0.30	3	1	1	0	1	0
2	0.06	11	10	10	11	11	0
2	0.15	5	2	3	4	4	1
2	0.30	0	0	0	0	0	0
Total		211	156	163	186	190	7

Across the acceptance batches, the additional direct accepts of the new envelope over the separate-gain envelope are: easy pool 0 (saturated), sensitivity 10, validation 1, heterogeneous links 7. The gain is thus batch-dependent and modest wherever positive; no distributional claim is made from correlated units.

7.5 Complementary certificates and staged screening

The $6 + \kappa$ quadratic bound and the six-scalar envelope use different information, and neither dominates. Taking the per user maximum and accepting a candidate only when every user passes the common threshold yields a strictly larger direct acceptance set: 424 of 472 feasible units (89.83%) in the exploratory grid versus 408 (86.44%) and 390 (82.63%); 139 of 159 (87.42%) versus 138 and 126 in the validation batch; and 190 of 211 (90.05%) versus 186 and 163 in the heterogeneous pool. The gain over the quadratic bound comes entirely from units certified only by the six-scalar envelope (16, 1 and 4 units respectively); no unit is accepted by the joint rule while both individual rules fail. The

separable bound is dominated by the envelope at every user problem in all pools and is dropped from the flow.

Table 7. Joint certification and selected pipeline (256 cells). Separate-gain = comparison envelope; confirmation entry: 202 is the common final accepted set; per-method entries are reference-call totals.

Batch	Feasible	Q_G	L_quad	Joint	Speedup	Calls
Exploratory (1080 units)	472	390	408	424	2.12× (25/30)	2946→731
Validation (540 units)	159	126	138	139	—	—
Realistic pool (540 units)	211	163	186	190	—	—
Confirmation (540, fresh)	202*	471	422	394	2.01× (11/15)	1424→394

Staged screening pipelines complete the same final decision task: bounds are evaluated only on users not yet certified, a failed bound is never a rejection, the tolerance-stopped exact reference ends a candidate at the first clearly infeasible user, and each user's factorization is cached. With one warm up and three interleaved repetitions, the ordering quadratic bound, then envelope, then reference — selected by a fixed rule on the exploratory cohort and frozen, with code hashes and unused seeds recorded before any new computation — attains a median paired speedup of 2.12 over the pure tolerance-stopped reference (25 of 30 pairs) on the exploratory grid and 2.01 (11 of 15) on five fresh confirmation seeds (221–225), reducing total reference calls by 75% and 72% respectively. The quadratic bound alone is the fastest single bound pipeline in the tested implementation; the envelope stage contributes the additional certified units and a further 7% reduction in reference calls, at some increase in pipeline time relative to the quadratic-only stage. Adding the envelope raises coverage; it does not, by itself, reduce wall-clock time. All pipelines produce identical final decisions in every run. An adaptive interval refinement (64→256→1024→4096 cells, stopping when a rounding guarded feasible point value shows the threshold cannot be crossed) reproduces fixed 4096 cell acceptance exactly with 3.9× fewer interval cells but is slower in wall clock at this scale and is not used.

A corrected interface study uses the same seeds 401–402 and ten-radius grid as the earlier interface study; the corrected protocol and code hashes were recorded before this rerun, which is not a fresh independent validation cohort. Six- and seven-scalar clients receive only their allowed summaries and external parameters. The compact reference receives only the spectral cache of Section 6.4. Batches visit all 48 users in each seed, with 1, 10 or 100 radius queries per user; actual invocation counts are respectively 48, 480 and 4800. Longer batches cycle the fixed radius grid. One warm-up and five interleaved repetitions rotate branch order. The reported per-query cost divides measured batch time by the instrumented invocation count, not by an assumed workload.

For the 100-query batches, median amortized costs are 44.24 µs for the six-scalar client, 0.25 µs for the seven-scalar quadratic client, and 547.30 µs for the compact reference. These are query-only measurements; the common threshold calculation is excluded here and included in the end-to-end flows. Float64 payloads are 48, 56 and 144 bytes respectively, excluding Python object overhead and common external parameters. Independently measured construction costs per user are 0.0459, 0.0469 and 0.0254 ms. The existing shared summary builder forms A and g even for the six-scalar branch; the timing describes this implementation, not the minimum possible cost of constructing six summaries. Compact construction is timed directly from the same raw inputs and is not charged the summary builder's work.

Direct coverage remains 119 for Q_G, 123 for L_{quad}, and 124 for their maximum over 240 candidate–radius units, of which 130 are reference-feasible: 91.5%, 94.6% and 95.4% feasible coverage. The joint rule adds exactly one unit over the quadratic certificate. Actual fallback now invokes the compact solver only for users not certified by the chosen summary bound, with fixed user order and early termination on a clearly infeasible user. Across both seeds, six-scalar, quadratic-only, joint and pure compact flows make 133, 126, 123 and 713 compact queries respectively. All final decisions agree with the full reference, with zero unresolved units in this pool. Median end-to-end query-flow

seconds per seed are 0.0549, 0.0375, 0.0511 and 0.2039; descriptor construction is separate. A client holds summaries and a service holds the richer spectral cache; these local measurements do not include network delay. The quadratic-only flow remains faster than the joint flow despite the latter's three fewer compact calls. Thus this study supports coverage supplementation, not a runtime advantage of the new envelope over the seven-scalar certificate.

Table 7a. Corrected interface study. Query cost uses 100 queries per user; calls sum both seeds; flow time is the median per seed. Construction is separate.

Client	Payload B	Query µs	Direct /240	TR calls	Flow s
Six scalars	48	44.24	119	133	0.0549
Seven scalars	56	0.25	123	126	0.0375
Joint rule	56	44.22	124	123	0.0511
Compact TR	144	547.30	exact	713	0.2039

The V6.4 interface timing normalization and simulated fallback counts are superseded and are not evidence for the revised claims. This correction does not change the historical certification pools or their theoretical definitions.

A controlled scaling study isolates dimension from load: four users, per base station power 1 (per stream 0.5), $\gamma = 1$, $\rho = 0.15$, identical candidate rules; only the antenna count varies (stacked dimension 4 to 32). The per-user-problem costs in Table 8 exclude summary construction, and the reference cost shown reuses each user's cached eigendecomposition; the pipeline seconds include summary construction. Interval bound evaluation is dimension independent (0.04–0.05 ms per user problem) because it operates on the summaries; the exact reference grows from 0.36 to 0.45 ms per user problem; the full certification flow is not dimension independent, since summaries and fallback scale with dimension. Relative to the tolerance-stopped pure reference, the median pipeline speedup is near parity at dimension 4 and 2.3–3.5× at dimensions 8–32. Four dimension points with changing channel geometry and feasibility do not establish a monotone speedup trend or a complexity order; acceptance counts are identical across pipelines at every dimension and are not compared across dimensions.

Table 8. Controlled scaling: dimension varies, all else fixed (4 users, per-BS power 1, γ = 1, ρ = 0.15; medians over three seeds). Per-user-problem ms exclude summary construction; the reference reuses each user cached eigendecomposition; pipeline seconds include summary construction. Four dimension points do not establish a complexity order.

Dim	Bound ms	Ref ms	Ref/Bound	Pure ref (s)	L_quad + ref (s)	Lq→Q_G + ref (s)
4	0.041	0.356	8.7×	0.0192	0.0193	0.0254
8	0.050	0.368	7.4×	0.0378	0.0165	0.0236
16	0.044	0.381	8.7×	0.0563	0.0175	0.0161
32	0.042	0.435	10.4×	0.0859	0.0310	0.0310

7.6 Independent exact cross-validation on the validation batch

A preregistered 74 case check set and the full 2160 problem validation batch are solved by the corrected SDP pipeline of Section 6.3. On this batch, the certified lower bound never exceeds the reference (maximum excess -1.2×10^{-13}), the witness upper bound agrees with the reference within a preset tolerance (largest observed deficit 1.3×10^{-12}, floating-point evaluation noise), the ordering $L \leq U$ holds with no violation, there is no false acceptance and no false rejection, all witnesses lie inside the ball (956 of 2160 required projection), and every user problem is decided (1247 feasible, 913 infeasible, zero unresolved) with witness quality $U - q^*$ of median 6.7×10^{-6} and maximum 1.6×10^{-4}. The maximum deviation of the raw SDP optimum from the reference is 5.6×10^{-8}. Within this batch, the two independent exact formulations are mutually consistent within the stated

tolerances; the formulations are exact, while all computed values are floating-point approximations. Other batches were not re-solved by the SDP and no cross-validation claim is made beyond this batch.

Table 9. S-procedure SDP baseline on the 2160-problem validation batch.

Quantity	Value
$L \le q^*$ (lower-bound safety), max excess	−1.2e-13 (never above)
U vs reference (witness validity)	agrees within tolerance; largest deficit 1.3e-12 (fp noise)
Ordering $L \le U$, violations	0
False acceptance / false rejection	0 / 0
Witnesses inside the ball	2160/2160 (956 projected)
Decisions	1247 feas., 913 infeas., 0 unresolved
Witness quality $U - q^*$ (913 cases)	median 6.7e-6, max 1.6e-4
Max $\lvert t - q^* \rvert$	5.6e-8
Per solve: build / solver / wall	0.33 / 1.6 / 5.7 ms

7.7 Timing and implementation sensitivity

Table 11 reports the same final decision task against the validated tolerance-stopped reference, which is the primary comparison. With the tolerance stop, the pure reference runs at 0.041 s per seed and threshold on the validation batch; among the two pipelines of Table 11, the new bound with tolerance-stopped fallback (0.032 s) is about 20% below it, while the same screening pipeline with the original 200 iteration fallback (0.075 s) is slower — part of the apparent saving in early measurements traced to multiplier search over iteration in the reference solver rather than to screening. The new pipeline is faster than the separate-gain pipeline in 6 of 15 validation pairs and 11–13 of 30 exploratory pairs across runs; no stable runtime advantage over the separate-gain envelope is observed. Table 10 reproduces the original 200 iteration protocol as a supplementary record. The results do not establish complexity advantages or superiority over an optimized solver. On the same batch, the generic SDP pipeline is 17.3 times slower than the tolerance-stopped reference as a ratio of medians (median per-pair slowdown 15.7, range 13.2–18.8; the reference is faster in 15 of 15 pairs), and the staged $L_{quad} \rightarrow Q_G \rightarrow$SDP pipeline is 6.5 times slower (median per-pair slowdown 7.2, range 1.5–12.4); these are single-round figures from the corrected SDP study, with all ratios defined as pipeline time divided by reference time. The SDP pipeline's value is cross validation, and the slowness of the generic conic path is not a statement about full-information methods.

Table 10. End-to-end seconds against the original 200-iteration reference (supplementary); total reference calls across ten seeds in parentheses.

γ	Reference	D + fallback	Old Q + fallback	Q_G + fallback	L_quad + fallback
0.5	0.1787 (1321)	0.0375 (244)	0.0313 (160)	0.0316 (147)	0.0248 (164)
1	0.1396 (1005)	0.0564 (379)	0.0516 (306)	0.0511 (297)	0.0429 (285)
2	0.0874 (620)	0.0607 (410)	0.0612 (384)	0.0640 (377)	0.0536 (353)

Table 11. Same final decision task against the validated tolerance-stopped reference (median seconds; total calls in parentheses).

Pipeline	200-iteration reference	Tolerance-stopped reference
Pure reference	0.1496 (1319)	0.0411 (1319)

Pipeline	200-iteration reference	Tolerance-stopped reference
New bound + fallback	0.0750 (496)	0.0323 (496)

8 Interface to configuration lifecycle management

The certificate addresses whether a fixed candidate can be deployed at a prescribed uncertainty radius. Once a candidate is selected, a separate complete configuration reference can determine its robust radius for subsequent online retain or refresh management. Candidate generation, deployment costs and channel drift remain outside the present problem. A positive certificate at one radius is not a maximum radius certificate.

9 Discussion

The sharpness statement is intentionally strong and narrow: it holds for the unrestricted six-scalar information class at fixed r, γ, σ^2 and dimensions; the approximating construction gives no uniform nominal-channel-norm bound over the class, divergence occurs in particular families, and bounded or attaining constructions exist in others (Remark after Theorem 2). Restricting the class — for example by adjoining $H = \| \hat{h} \|$, for which (28) gives a necessary compatibility condition — can only shrink the admissible set, and the six-scalar envelope need not remain sharp; the complete envelopes of the H-branch and of other richer interfaces are open. Empirically, the certificate gain over the separate-gain envelope is positive in the sensitivity, validation and heterogeneous-link batches and zero in the saturated easy batch; the corrected strict-client study finds the seven-scalar certificate cheaper than the six-scalar envelope, while joint screening adds one feasible unit and reduces compact calls at higher wall-clock cost than quadratic-only screening. The compact value cache reduces the numeric payload comparison to 48 versus 144 bytes at dimension eight; this factor is dimension-dependent. The engineering motivation is a fixed-size information interface, not a demonstrated advantage over the seven-scalar alternative. Exact fallback guards remain floating-point rather than formally validated intervals. The certificate is complementary to, not a replacement for, full-information methods: within the branched family of interfaces it is the strongest guarantee available at its information level, and combining interfaces by maximization is safe.

10 Conclusion

Mathematically, joint spectral compatibility of three gains of one interference operator yields an explicit three-branch directional extremum, a safe shared-budget QoS envelope dominating the separate-gain envelope, equality with the information-class infimum away from the degenerate boundary together with an explicit approximating construction, and a separately derived, explicitly attained phase-coupled envelope at that boundary. Practically, the envelope is a fixed-size certificate whose empirical gain over the separate-gain envelope is batch-dependent and modest, and which is complementary to a quadratic bound retaining one additional scalar — the two have no universal dominance relation and combine safely by maximization, giving the highest direct-certification coverage among the tested bounds. In scope, the certificate does not replace exact full-information solvers, does not optimize candidate generation, and its sharpness concerns the stated unrestricted information class; natural extensions include richer retained summaries such as the H-branch, physically restricted information classes, validated numerical certificates, and lifecycle management of stored configuration certificates.

Appendix A: formula-level comparison with the closest literature

Table 12. Formula-level comparison: what is directly used, what is context, what is evaluated here.

Ref	Original problem / assumptions	Original conclusion	Direct substitution yields our result?	Remaining steps
[5] Buzano; Bottazzi–Conde	inner-product bounds for |<Tx,y>| type quantities	two-sided C-S refinements	No: cross-term bound (4) proved by our 2x2 determinant argument	conceptual kinship only
[6] Grone et al.	partial Hermitian matrix, chordal specified graph	PSD completion iff specified blocks PSD	No: our 2x2 completion verified directly	general theorem in background
[7] Parrott	contraction completion / quotient norm, lifting	operator-ball structure of completions	No: lifting theorem not applied	completion philosophy cited as context
[8] Micchelli–Rivlin	optimal recovery from finite data	radius/center of information theory	No: B*(s) is an infimum of margins, not a recovery-error radius	viewpoint relation; no equivalence claimed
[9] Kim et al.	worst-case SINR, norm-bounded steering error	parallel/orthogonal error decomposition bounds	No: min{b0 r, a x + b y} not found as a stated envelope there	decomposition context; baseline assembled here

The searches covered the operator-inequality, completion, optimal-recovery and robust-beamforming lines as documented in the project audit trail; within those searches we did not locate the closed form (6) under exactly the three retained summaries and a shared budget, nor the boundary envelope (25). Absence of a match in these searches is not a proof of novelty.

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Reproducibility

The supplied code reproduces every number. `run_experiments.py` (original pools), `audit_and_stress.py` (sensitivity study), `run_v41.py` (preregistered validation batch and theory checks), `run_optimization.py` (joint certification, staged screening, adaptive refinement, scaling), `run_sdp_v2.py` (corrected SDP baseline), `experiments_v6.py` (heterogeneous-link pool and controlled scaling) and `v61_data_audit.py` (cross-batch re-aggregation from raw CSVs) write into separate immutable result directories and exit nonzero on failed checks. An earlier SDP implementation with an invalid negative-lower-bound rejection rule is retained as history with correction headers and is not cited. Numerical minima are diagnostics; interval lower bounds are used for acceptance.

The V6.5 correction is reproduced by `run_v65.py` and `v65_clients.py`, writing configuration hashes, validation records, measured preprocessing and query batches, actual fallback invocations and summaries into `results_v65/`. V6.4 interface performance figures are retained only as superseded history.